

Encoder: shorten the sequence

Input window
1 x 512



Encoder 1: residual conv
32 x 256



Encoder 2: residual conv
64 x 128



Encoder 3: residual conv
128 x 64



Encoder 4: residual conv
256 x 32



Dilated bottleneck: 256 x 32
Dilation rates: 1, 2, 4, 8

Decoder: reconstruct the ECG

Linear output: 1 x 256
Resize to 1 x 512



Decoder 4: up block
32 x 256



Decoder 3: up block
32 x 128



Decoder 2: up block
64 x 64



Decoder 1: up block
128 x 32



----- skip x 0.5 ----->

----- skip x 0.5 ----->

----- skip x 0.5 ----->

----- skip x 0.5 ----->

Dimensions are channels x time samples; the batch dimension is omitted.

Dashed arrows carry encoder features to matching decoder lengths.

Each up block resizes to its skip length; final interpolation restores 512 samples.